

MD RAKIBUL ISLAM RAIHAN

AI Engineer - LLM Inference & Multimodal Research (2027 New Grad)

Xi'an, China | Open to Relocation | Available July 2027

Nationality: Bangladesh | Sex: Male | DOB: 2002.12.06

Languages: Bengali (Native) | English (Fluent) | Chinese (Conversational)

Work Authorization: Requires China work visa (employer-sponsored Z Visa) | Global: Sponsorship welcome

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Portfolio: theraihanrakibb.github.io/Online-Portfolio



Dear Sir or Madam,

I am writing to apply for an AI Engineer position focused on LLM inference and multimodal research (2027 New Grad). I am an M.Eng. Software Engineering candidate at Northwestern Polytechnical University (Top 1%, 985/211, Chinese Government Scholarship), graduating July 2027. My work sits at the intersection of efficient multimodal generation and the systems that make such models deployable, and I am seeking a role - in an industrial research group or as a research engineer - where publication-quality work and production systems depth are both valued.

Research. I am first author of an IEEE TNNLS (Q1) paper under review: NVFP4-DiT, a 4-bit audio-guided video diffusion transformer achieving 4x memory reduction while maintaining generation quality. The work combines low-precision quantization (FP8/FP4/NVFP4) with multimodal cross-attention, and I benchmarked it on A800/H100 GPUs using PyTorch Profiler and Nsight Systems. My M.Eng. thesis focuses on multimodal deepfake detection using visual, audio, and temporal consistency analysis, and my B.Eng. thesis designed a distributed confidential-query framework using Apache Spark and CKKS homomorphic encryption - achieving 95% privacy preservation and a 20% performance improvement.

Research Interests. Efficient multimodal generation and understanding; low-precision (FP4/FP8/NVFP4) training and inference; and co-design of models with the serving stack - KV-cache management, PD disaggregation, and RDMA transport. I am drawn to problems where a modeling decision and a systems constraint must be solved together, and I would be glad to bring both my first-author track record and my production experience to a research group.

Systems. During my internship at InfiX.ai, I built production LLM serving on multi-node A800 and H800 GPU clusters using SGLang. I designed a 3-tier KV-cache offload pipeline (GPU HBM -> CPU DRAM -> SSD) over Mooncake RDMA, achieving a 92.27% cache hit rate and an 8.3x TTFT reduction (55s -> 6.7s) for long-context models. I patched SGLang source directly (kvcacheio.py, hiradix_cache.py, memory_pool_host.py) to fix production cache failures and deploy hybrid-attention Qwen models. I also built PD-disaggregated (prefill/decode) serving with RDMA KV sharing, raising stable concurrency from 192 to 256 requests, and profiled CUDA/Triton inference with Nsight Systems to localize GEMM bottlenecks and apply FP8 quantization for throughput gains. Beyond that, I diagnosed and resolved 12+ production incidents across RDMA, etcd, networking, storage, and distributed inference.

Open Source. I maintain ai-infra-portfolio, a curated suite of 10 production-grade, Dockerized AI-infra projects: LLM serving, observability, distributed training, MLOps, and API gateways; train-launcher, a CLI that launches and supervises multi-node PyTorch training with fault tolerance and automatic retry; and c-compiler-frontend, a self-built C compiler (lexer/parser/codegen) using Yacc/Lex.

Stack. Python (Proficient) | C/C++ | Bash | SQL | SGLang | KV-cache & PD-disaggregation | Mooncake RDMA | FP8/FP4 quantization | CUDA | Triton | multi-GPU serving (4x A800) | PyTorch | Apache Spark | CKKS | Linux | Docker | Git | CI/CD.

What makes my profile unusual is the combination of algorithmic research and production systems depth - I understand both the modeling side and the hardware and serving constraints that make efficient multimodal AI necessary and hard. This lets me move fluidly between research experiments and deployed infrastructure. I would welcome the chance to discuss how my experience can contribute to your team or research group; my CV, NVFP4-DiT manuscript, and thesis are available on request. Thank you for your consideration.

Sincerely,

MD RAKIBUL ISLAM RAIHAN

M.Eng. Software Engineering, NWPUP (Expected 2027)